

LESSON 11.4

FEATURES OF PROPORTIONAL RELATIONSHIPS



LESSON INTRODUCTION

MA.7.AR.4.1 Determine whether two quantities have a proportional relationship by examining a table, graph, or written description.

LEARNING TARGETS

- I can determine if two quantities have a proportional relationship.
- I can relate tables and graphs that represent the same relationship between two quantities in a real-world context.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.8.AR.3.1 Determine if a linear relationship is also a proportional relationship.

ESSENTIAL QUESTIONS

- How do you determine if two quantities have a proportional relationship?
- How can you use a table or graph to represent a proportional relationship?

LESSON NARRATIVE

Students will be introduced to using tables and graphs to determine if a relationship is proportional or not. Students relate different representations of relationships, including written descriptions, tables, and graphs, before determining if the relationship is proportional. Students learn to determine whether a relationship is proportional from a table or graph, including explaining how they know.

COMPONENT SUMMARY

11.4.1 WARM-UP (~5 MINS)

Notice and wonder with graphs of two different relationships

11.4.3 BRING IT TOGETHER (5-10 MINS)

Synthesize how to determine if a relationship is proportional

11.4.5 WRAP-UP (5 MINS)

Explaining proportional relationships

11.4.2 ACTIVITY (15-20 MINS)

Relate written descriptions of proportional relationships to graphs and tables of values

11.4.4 YOUR TURN! (~5-10 MINS)

Identifying whether relationships are proportional

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- Cards for card sort

ADDITIONAL PREPARATION

- Consider using the Five Practices for Orchestrating Productive Class Discussion when planning for the 11.4.2 Activity and 11.4.3 Bring It Together.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to recognize proportional relationships from graphs or tables.
- Students may not recognize how to use the ordered pairs from a table of values to determine if a relationship is proportional.

THINGS TO CONSIDER

- Consider having students make one stack of cards from 11.4.2 Activity that shows proportional relationships and another stack of cards that does not show proportional relationships. Work with students to either create a T-chart or a Venn diagram noting the characteristics of each group of cards.

LITERACY AND LANGUAGE ROUTINES

Clarify, Critique, Correct

11.4.2 Activity and 11.4.4 Your Turn! require students to provide analysis and explanation of their responses. Allow students to swap their responses with another student. Students will then ask their partner for clarification and additional analysis as needed. This process allows students to think more analytically and to provide more detailed mathematical explanations.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

11.4.2 Activity allows students to use visual and mathematical patterns to identify common features of proportional relationships. Students analyze the tables and graphs to look for evidence of a constant ratio between each coordinate pair. Support students as they use their observations of patterns to generalize about proportional relationships.

DIFFERENTIATE INSTRUCTION

Read and discuss the problems and scenarios aloud to assist students who benefit from auditory entry points. Emphasize the visual nature of the proportions for students who prefer visual learning methods.

Consider using the Five Practices for Orchestrating Productive Class Discussion to plan for how to facilitate a discussion that highlights various students' voices and understanding as they further explore these relationships.

11.4.1 WARM-UP: NOTICE AND WONDER

TEACHER MOVES

Notice and Wonder

Engage students in a Notice and Wonder routine to draw conclusions about the cost of peaches and plums based on the graph. After students complete the Warm-Up independently, consider asking questions like:

- How would you describe the cost-to-pound ratio of plums/peaches?
- Why do you think cost is shown on the y-axis?
- What does this tell you about the relationship of the cost of an item to its weight?

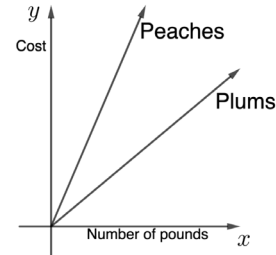
QUESTIONING

If time allows during the debrief, suggest students mark where $x=1$ could be along the x-axis and then discuss with a partner what they notice about the y-values when $x=1$. Ask if they all had to use the same location for $x=1$ for this relationship to be true.



11.4.1 Warm-Up: Notice and Wonder

The graphs below show the cost, y , of buying x pounds of fruit. One graph shows the cost of buying x pounds of peaches, and the other shows the cost of buying x pounds of plums.



1. After looking at the relationships graphed, complete the table with one thing you notice and one thing you wonder.

One thing I notice is...	One thing I wonder is...
<i>Answers will vary.</i>	<i>Answers will vary.</i>

2. Explain which fruit you think costs more per pound.
Peaches cost more per pound because the line is steeper, meaning the price for 1 pound of peaches is higher on the y-axis than for 1 pound of plums.



11.4.2 Activity: Card Sort

Several scenarios are described in the chart. Your teacher will give you two sets of cards showing tables and graphs.

- Work with your partner to match a table of values and graph to each scenario. Write the matching table number and letter of the graph in the chart.

Written Description	Table of Values	Graph
a. T-shirts are priced \$10 each at a store. There is a sale posted that says, "Buy two T-shirts, get the second one half-price." The table and graph show the total cost, y , of x T-shirts.	1	C
b. A movie streaming service costs \$8.49 each month. The table and graph show the total cost, y , for x months of the subscription.	2	A
c. A recipe uses 2 tablespoons of honey, x , for every 8 cups of flour, y .	5	F
d. Camping fees, y , at Blue Springs State Park in Volusia, FL include a \$6.70 reservation fee and \$24 per night, x .	3	E
e. Rachel ran 6 laps in 18 minutes around the track at Riverview High School in Sarasota, FL. She notes the average number of minutes, y , that have passed after each lap, x .	4	B
f. Kenneth is filling his fish tank with water at a flow rate of 1.6 gallons per minute. The table and graph show the number of gallons of water, y , after x minutes.	6	D

- Complete the statements using the words from the bank.

constant	equivalent	line
ordered	origin	table

When two quantities are in a proportional relationship, the graph will include the origin, $(0, 0)$ and the points on the graph will form a straight line. The ordered

pairs of the points on the graph match the pairs of values in the table representing the same relationship. The ordered pairs in a proportional relationship are equivalent ratios. The value of the ratio is called the constant of proportionality.

- Work with your partner to determine which scenarios from the card sort represent proportional relationships using the matching tables and graphs. Write the letter P next to the written description in the chart for each relationship that is proportional.
Labeled in chart. The written descriptions for b, c, e, and f represent proportional relationships.
- Use Graph D and the context it represents to complete the following parts with your partner.
 - Discuss why, for this context, it makes sense that every point on the line shown is possible.
 - Explain whether you think the graph representing this relationship will continue increasing every minute that passes.
The relationship should not continue increasing because a fish tank can only hold a set amount of water. Eventually, it would overflow.

11.4.2 ACTIVITY: CARD SORT

TEACHER MOVES

This activity is scaffolded to allow students to engage with it with their partner rather than be guided through by the teacher. There will be an opportunity to summarize and synthesize in the next component. Students may revise their thinking throughout the activity as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group for 11.4.3 Bring It Together. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed activity.

TEACHER MOVES

MA.K12.MTR.5.1: Patterns and Structures

Support students as they analyze the tables and graphs by encouraging them to use their observations of patterns to form generalizations about proportional relationships.

TEACHER MOVES

Question 4 and its parts provide an opportunity for the teacher to listen to students' discussions, to observe whether they understand why the graph is continuous, and if they make connections to constraints on real-world contexts, which they explored in Unit 10. The intention of this question is to provide students with an opportunity to discuss and share their current thinking, not necessarily that they will all get this correct at this point.

The teacher can then use these observations from listening to the discussions about this question (along with the previous ones) to determine which areas may need more emphasis in the remainder of this lesson and Lesson 5, as students continue working on this benchmark.

**Note: It is not expected that students use the vocabulary of "continuous/discrete" or "constraint." Students can use their own words to describe these concepts as they consider the real-world context.*

11.4.2 ACTIVITY: CARD SORT (CONTINUED)

ENRICHMENT

If students are struggling with question 5, consider asking:

- If a relationship is proportional, what do we know about the ratios formed from each pair of values? (They are equivalent. (Point back to question 2.))

If students are still stuck, ask:

- If the ratios of each pair of values are equivalent for a proportional relationship, how will the ratios compare to each other if the relationship is not proportional? (They will not be equivalent.)

If the majority of the class is struggling with this, consider going over this question before or after the 11.4.3 Bring It Together.

5. Use Graph E and the table it matches to complete the following parts with your partner.
 - a. The points on Graph E form a straight line, but the situation does not represent a proportional relationship. Explain why it does not represent a proportional relationship.
The ordered pairs in the table do not form equivalent ratios. Also, the line does not pass through the origin.
 - b. Use at least two ordered pairs from the table of values that matches Graph E to show why this relationship is not proportional.

Answers will vary.

$$\frac{30.7}{1} = 30.7 \neq \frac{78.7}{3} = 26.2\bar{3} \neq \frac{102.7}{4} = 25.675$$

11.4.3 BRING IT TOGETHER: DETERMINING IF A RELATIONSHIP IS PROPORTIONAL

TEACHER MOVES

This component is intended to be teacher-led and to summarize the key take-aways from the activity in 11.4.2 Activity.

Consider using observations from the activity to determine which key ideas need to be solidified or clarified. This could include discussing specific questions the student worked on. It should not be the teacher going through each question and telling the correct answer. Instead, the teacher might consider calling on students to share their thinking and respond to one another or restating something they heard while observing and asking students to discuss as the teacher leads the conversation toward solidifying these ideas.

Consider incorporating the Five Practices for Orchestrating Productive Class Discussion.



11.4.3 Bring it Together: Determining if a Relationship is Proportional



Graphs representing proportional relationships have the following in common.

- The points that satisfy the relationship lie on a straight line.
- The line they lie on passes through the origin, (0,0).



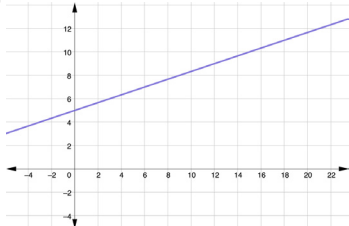
Tables representing proportional relationships have the following in common.

- Each row in the table shows a ratio of the two quantities in the relationship.
- All of the ratios in the relationship are equivalent.



11.4.4 Your Turn!

1. A graph is shown.



- a. Complete the statement.

The graph ☐ represents
☒ **does not represent** a proportional relationship between two quantities.

- b. Explain your selection for Part A.

The line does not pass through the origin.

2. Fountain drinks at Wawa are priced by size. The volume in fluid ounces (fl. oz.) and price in dollars (\$), before tax, for each size is shown in the table.

Volume (fl. oz.)	16	21	30
Cost (\$)	1.49	1.59	1.89

- a. Complete the statement.

The table ☐ represents
☒ **does not represent** a proportional relationship between two quantities.

- b. Explain your selection for Part A.

The ratios of the two quantities from the table are not equivalent.

11.4.4 YOUR TURN!

TEACHER MOVES

Clarify, Critique, Correct

Consider having students analyze the responses of a partner. Students will then ask their partner for clarification and additional analysis as needed. This process allows students to think more analytically and to provide more detailed mathematical explanations.



11.4.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to help them understand how to determine if two quantities have a proportional relationship.

Two quantities have a proportional relationship if:

#1 – the line passes through the origin and

#2 – the ratios formed from the ordered

pairs, $\frac{y}{x}$, are all equivalent.

11.4.5 WRAP-UP: COOL DOWN

DIFFERENTIATE INSTRUCTION

Allow students to complete the Wrap-Up verbally if they struggle to explain their thoughts in writing. Students may also prefer to draw and label an example to demonstrate a proportional relationship.

DIFFERENTIATE INSTRUCTION

Use the information provided by students to inform future instructional approaches as well as decisions about differentiating instruction in future lessons.